

M731–Partial Differential Equations HOMEWORK – Fall 2026

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All homework should be uploaded in Canvas by the due date and time.

SET 1: Due Thursday 09/24/2026 by 7pm

From McOwen's Book: read the introduction & Sections 1.1d) and 1.3

From McOwen's Book Section 1.1 do: 1, 2, 3, 4a)b), 6a) 9.

SET 2. Due: Thursday 10/8/2026 by 7pm

Additional Problems. Read carefully the first Handout posted on the course webpage about inhomogeneous transport equations. You should be familiar with this from undergraduate PDE at least when $d = 1$. Here this is in any dimension $d \geq 1$. Note that in the Handout I wrote n instead of d for the dimension so adjust the notation to the additional problems 1 and 2 below.

(1) Write down an explicit formula for a function u solving the inhomogeneous initial value problem

$$\begin{cases} u_t + b \cdot \nabla u = f & \text{on } \mathbb{R}^d \times (0, \infty) \\ u = g & \text{on } \mathbb{R}^d \times \{t = 0\} \end{cases}$$

where $f = f(x, t)$, $f : \mathbb{R}^d \times [0, \infty) \rightarrow \mathbb{R}$, $c \in \mathbb{R}$, $b \in \mathbb{R}^d$ and $g : \mathbb{R}^d \rightarrow \mathbb{R}$ are all given. (Hint. Solve first the homogeneous problem (i.e. $f = 0$) using the method of characteristics (as in class or McOwen section 1.1a)). Then use the Fundamental Theorem of Calculus to write $u(x, t) - g(x - bt)$ for (x, t) along a characteristic ($g(x - bt)$ is the homogeneous solution) as an appropriate integral of f . At the end u will be the sum of the homogeneous solution plus an integral term.)

(2) Write down an explicit formula for a function u solving the initial value problem

$$\begin{cases} u_t + b \cdot \nabla u + c u = 0 & \text{on } \mathbb{R}^d \times (0, \infty) \\ u = g & \text{on } \mathbb{R}^d \times \{t = 0\} \end{cases}$$

where $c \in \mathbb{R}$, $b \in \mathbb{R}^d$ and $g : \mathbb{R}^d \rightarrow \mathbb{R}$ are given.

This problem below is a basic problem from undergraduate Analysis. Might be useful to think about it for balls in $\mathbb{R}^d$ first to gain intuition.

(3) Let f be a continuous function on an open set $D \subset \mathbb{R}^d$ such that

$$\int_{D_0} f(x) dx = 0 \quad \text{for all } D_0 \subset D.$$

Prove that then $f \equiv 0$ on D .

From McOwen's Book Section 2.1: 1, 2, 7

Note: I. will do related problem to 7 in the first half of class on Tuesday 10/06 to show an example failure of stability -continuous dependence of the solution on its data)

From McOwen's Book: Read carefully **McOwen Section 2.2a) pages 49-51** classifying equations of second order in $\mathbb{R}^2$. You have learned this classification (hyperbolic, parabolic, elliptic) in undergraduate PDE but here the conditions are expressed in terms of characteristic curves. After reading carefully, rewrite this section yourselves filling any gaps and explaining the steps and conclusions. Write or type clearly your notes and hand them to me on Thursday.

SET 3. Due: Thursday 10/15/2026

From McOwen's Book Section 2.2: Read carefully all. of Section 2.2b). Then **do** 1.

Additional Problems.

1) a) Find the characteristics of the PDE

$$y^2 u_{xx} - 2y u_{xy} + u_{yy} = u_x + 6y,$$

and determine if elliptic, parabolic or hyperbolic.

b) Then find the canonical form and use it to find the solution u first in the ξ and η variables and then in the x and y variables.

2) a) Find the characteristics of the PDE

$$x u_{xx} + (x - y) u_{xy} - y u_{yy} = 0, \quad x > 0, y > 0,$$

and determine if elliptic, parabolic or hyperbolic.

b) Then show that it can be transformed into the canonical form

$$(\xi^2 + 4\eta) u_{\xi\eta} + \xi u_\eta = 0$$

for ξ and η are suitably chosen canonical coordinates. and use this to obtain the general solution in the ξ and η variables.